
AMPLIFAI: A Multiphase CT Dataset for Benchmarking Clinical Reasoning in LI-RADS Assessment of Liver Lesions

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Abstract

Hepatocellular carcinoma (HCC) is the third leading cause of cancer-related mortality worldwide, with early detection improving survival from <20% to >70%. The standardized LI-RADS criteria establish a biopsy-free, fully imaging-based framework that can serve as a foundation for automating HCC diagnosis with artificial intelligence (AI). However, the lack of large, publicly available datasets with high-quality labels has limited the development of AI models for LI-RADS characterization. We introduce the **AMPLIFAI** dataset, the first public dataset of multiphase abdominal CT scans annotated with LI-RADS categories and segmented for three major LI-RADS features: arterial phase hyperenhancement, washout, and enhancing capsule. Following the *Datasheets for Datasets* format, this paper details the dataset's composition, curation process, and annotation pipeline to facilitate transparent, reproducible research.

1 Introduction

Hepatocellular carcinoma (HCC) is the most common primary liver cancer and the third leading cause of cancer-related mortality worldwide, with early detection improving patient survival from <20% up to >70% [1, 2]. The standardized Liver Imaging Reporting and Data System (LI-RADS) [3, 4] defines the diagnostic criteria for clinical decision-making, relying on abdominal CT as one of the primary imaging modalities for diagnosis. Specifically, LI-RADS category 5 (LR-5) is considered "definitely HCC," [5–7] establishing a biopsy-free, fully imaging-based diagnostic pathway where artificial intelligence (AI) can enable early detection and improve patient outcomes [8].

Unlike many medical imaging classification tasks, HCC diagnosis on abdominal CT requires multiphase acquisition to evaluate dynamic contrast enhancement patterns [3]. This complexity makes data curation time-consuming, as scans need to be spatially registered across multiple phases and annotated for both phase-specific lesion characteristics and cross-phase feature interactions. As a result, there is a distinct lack of large, publicly available datasets with granular, feature-level annotations of major LI-RADS features required for developing robust AI models for HCC detection and characterization.

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We introduce the **AMPLIFAI** (**A**nnotated **M**ulti**P**hase **L**iver **I**maging **F**or **AI**) dataset, a standardized benchmark for AI-based characterization of HCC lesions in multiphase abdominal CT. The dataset consists of 590 cases curated and harmonized from four public datasets [9–12]. Expert annotations, provided by a team of five board-certified radiologists and one resident, include LI-RADS categories (LR-1 through LR-5, LR-TIV, and LR-M), lesion size, and voxel-level segmentation masks for three major imaging features: non-rim arterial phase hyperenhancement (APHE), non-peripheral washout, and enhancing capsule. In this paper, we provide structured documentation for AMPLIFAI following the *Datasheets for Datasets* format [13]. By detailing our dataset composition, curation process, and annotation pipeline, we aim to facilitate transparent, reproducible research in medical imaging AI.

This datasheet only covers the publicly released training and validation data. We do not intend to release the held-out private institutional test set nor provide additional details regarding it to ensure the integrity of the challenge. The primary sources used to compile this datasheet include the accepted AMPLIFAI [Challenge Proposal](#) and [Official Website](#) as part of the MICCAI 2026 Satellite Events.

2 AMPLIFAI Datasheet

2.1 Motivation

For what purpose was the dataset created? Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description.

AMPLIFAI focuses on the automated characterization of HCC lesions in multiphase abdominal CT using the standardized LI-RADS criteria. On a scale from LR-1 to LR-5, LI-RADS categories describe the increasing probability of HCC based on lesion morphology and dynamic contrast enhancement patterns. Specifically, LR-1 indicates a lesion is "definitely benign", while LR-5 is considered "definitely HCC" and does not require further tissue biopsy for diagnosis. This framework represents a fully imaging-based diagnostic pathway, where AI models have the potential to enable early diagnosis and improve patient outcomes.

The LI-RADS criteria consist of temporal analysis in multiphase CT, where multiple contrast phases are acquired: before contrast injection (non-contrast) and at successive time points after injection (arterial, portal venous, and delayed) to capture dynamic enhancement patterns in the lesion. The primary gating feature is APHE, where the lesion appears brighter than the surrounding healthy tissue in the early arterial phase. It is considered "non-rim" if the enhancement spreads internally and is not limited to the outer edge of the lesion. Only lesions exhibiting unequivocal non-rim APHE are considered for the highest probability categories (LR-3 to LR-5). However, if the enhancement is limited to the outer edge, the lesion indicates rim APHE and cannot be considered for LR-5.

If APHE is present, the LI-RADS category is determined by a combination of the lesion's size and how APHE interacts with two major features: First, non-peripheral washout, where the lesion becomes darker than the liver in the portal venous or delayed phases. Second, enhancing capsule, where a thin, peripheral rim remains enhanced in the portal venous or delayed phases due to contrast retention. The lesion size and presence of one or both features increases the probability for HCC. For example, a lesion with non-RIM APHE that is 2.3cm and has one additional feature (washout or capsule) is categorized as LR-5 ("definitely HCC") while a lesion without non-rim APHE can only reach a maximum of LR-4 ("probable HCC"), even if it is large and shows washout.

This presents a unique challenge for automating HCC diagnosis, as AI models must capture the spatio-temporal relationships between lesion morphology and dynamic enhancement patterns across multiple contrast phases. However, there are no large, publicly available datasets with granular, radiologist-annotated labels for major LI-RADS features. AMPLIFAI addresses this gap as the first public dataset to provide voxel-level segmentations for the three major imaging features (APHE, washout, and enhancing capsule). Annotated by a team of five board-certified radiologists and a resident, our dataset establishes a high-quality, reliable ground-truth for training robust AI models.

Who created the dataset (e.g., which team, research group) and on behalf of which entity (e.g., company, institution, organization)?

AMPLIFAI was created by the [Center for Applied AI](#) team at the [University of Maryland Institute for Health Computing \(UM-IHC\)](#). Our interdisciplinary team includes researchers, faculty, and clinicians from the University of Maryland, College Park, the University of Maryland School of Medicine, the

Table 1: Distribution of cases across source datasets for training and validation splits.

Source	Training	Validation	Total
PLC-CECT [12]	277 (52.17%)	29 (49.15%)	306 (51.86%)
WAW-TACE [10]	176 (33.15%)	23 (38.98%)	199 (33.73%)
HCC-TACE-SEG [11]	41 (7.72%)	4 (6.78%)	45 (7.63%)
TCGA-LIHC [9]	37 (6.97%)	3 (5.08%)	40 (6.78%)
Total	531 (100%)	59 (100%)	590 (100%)

University of Maryland Medical System, and the U.S. Food and Drug Administration. The complete list of authors is provided on the title page.

Who funded the creation of the dataset? If there is an associated grant, please provide the name of the grantor and the grant name and number.

The UM-IHC Center for Applied AI team is supported in part by the Montgomery County, Maryland and the University of Maryland Strategic Partnership: MPowering the State, a formal collaboration between the University of Maryland, College Park and the University of Maryland, Baltimore. We also acknowledge the support of the 2026 Accelerated Translational Incubator Pilot (ATIP) Grant Program from the University of Maryland, Baltimore, Institute for Clinical and Translational Research (ICTR), which is funded in part by the National Center for Advancing Translational Sciences (NCATS) Clinical Translational Science Award (CTSA), UM1TR004926. FXD is supported in part by a grant NIH CTSA 1K12TR004925-01A1. The contents of this manuscript are solely the responsibility of the authors and do not necessarily represent the official view of the National Institutes of Health.

Any other comments?

No.

2.2 Composition

What do the instances that comprise the dataset represent (e.g., documents, photos, people, countries)? Are there multiple types of instances (e.g., movies, users, and ratings; people and interactions between them; nodes and edges)? Please provide a description.

Each instance in the dataset represents a single multiphase abdominal CT study of a patient evaluated for HCC using the standardized LI-RADS criteria.

How many instances are there in total (of each type, if appropriate)?

The dataset contains 590 cases from 584 unique patients, derived from four public datasets: TCGA-LIHC [9], WAW-TACE [10], HCC-TACE-SEG [11], and PLC-CECT [12]. Most instances correspond to a single patient, with the exception of TCGA-LIHC, which includes multiple visits (up to 3) for 5 patients. To prevent data leakage, we split the dataset into a training set of 531 cases (90%) and validation set of 59 cases (10%) without any patient overlap. Uniform sampling was used across the source datasets to ensure a balanced data distribution in both sets. Table 1 shows the total number of cases in the training and validation sets, stratified by their source datasets.

The majority of the dataset ($> 85\%$) is derived from PLC-CECT and WAW-TACE. These well-documented and standardized datasets have large sample sizes and require minimal preprocessing and harmonization. In contrast, the HCC-TACE-SEG and TCGA-LIHC datasets have greater heterogeneity, requiring extensive preprocessing prior to inclusion. Section 2.3 provides a detailed description of our data curation pipeline.

Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set? If the dataset is a sample, then what is the larger set? Is the sample representative of the larger set (e.g., geographic coverage)? If so, please describe how this representativeness was validated/verified. If it is not representative of the larger set, please describe why not (e.g., to cover a more diverse range of instances, because instances were withheld or unavailable).

The dataset represents a non-random sample of studies from four public datasets: TCGA-LIHC was collected at the Mayo Clinic, the University of North Carolina, Alberta Health Services, and Lahey

Table 2: Distribution of contrast phases across training and validation splits.

Phase	Non-contrast	Arterial	Portal Venous	Delayed
Training	381 (71.75%)	531 (100%)	527 (99.25%)	406 (76.46%)
Validation	43 (72.88%)	59 (100%)	59 (100%)	45 (76.27%)
Total	424 (71.86%)	590 (100%)	586 (99.32%)	451 (76.44%)

Hospital and Medical Center [9]; WAW-TACE was collected at the Medical University of Warsaw between May 2016 and April 2021 [10]; HCC-TACE-SEG was collected at the University of Texas MD Anderson Cancer Center between November 2002 and June 2012 [11]; and PLC-CECT was collected at the Chongqing Yubei District People’s Hospital between January 2015 and December 2022 [12]. We only included multiphase abdominal CT studies of patients prior to HCC treatment in this analysis. Please refer to each source dataset’s publication for further details.

What data does each instance consist of? “Raw” data (e.g., unprocessed text or images) or features? In either case, please provide a description.

Each instance consists of a multiphase contrast-enhanced abdominal CT study, containing one arterial and at least one late contrast phase (portal venous and/or delayed). Non-contrast scans are included when available. Table 2 shows the distribution of contrast phases in the dataset. We provide all CT volumes and segmentation masks in the NIfTI format for accessibility. The total dataset size is about 146 GB.

Is there a label or target associated with each instance? If so, please provide a description.

Each case contains annotations for the LI-RADS category, lesion size (in mm), and the presence of three major LI-RADS features: non-rim APHE, non-peripheral washout, and enhancing capsule. Since patients may present with multiple liver lesions, we define the target lesion for LI-RADS characterization per case. For this target lesion, we provide voxel-level segmentation masks for both the overall lesion and the three major LI-RADS features. Importantly, since washout and enhancing capsule may appear in either or both late-contrast phases, we explicitly provide segmentation masks for these features across all phases in which they are present. All segmentations are provided as NIfTI volumes.

The dataset includes cases with normal livers (i.e., no liver lesions) as well as lesions annotated with one of seven standardized LI-RADS categories [3]:

- LR-1 ("Definitely benign"): Lesions show no enhancement in any contrast phase.
- LR-2 ("Probably benign"): Small, distinct nodules without the presence of major features.
- LR-3 ("Intermediate probability"): Lesions with equivocal presence of major features.
- LR-4 ("Probably HCC"): Highly suspicious lesions depending on their size and the presence of some major features.
- LR-5 ("Definitely HCC"): Large lesions with unequivocal presence of major features.
- LR-TIV ("Tumor in vein"): Unequivocal presence of soft tissue within a vein, regardless of whether an associated lesion is observed.
- LR-M ("Malignant"): Lesions that appear malignant but lack the typical characteristics of HCC.

The lesion size is measured as the longest dimension (in mm) of the target lesion in the axial plane. The presence of major LI-RADS features is provided as a binary label (i.e., 0/1), with the exception of APHE, which provides labels that distinguish non-rim APHE, rim APHE, and absence of APHE. All labels are included in the metadata CSV file of the training and validation splits.

Is any information missing from individual instances? If so, please provide a description, explaining why this information is missing (e.g., because it was unavailable). This does not include intentionally removed information, but might include, e.g., redacted text.

All instances are complete.

Are relationships between individual instances made explicit (e.g., users’ movie ratings, social network links)? If so, please describe how these relationships are made explicit.

There are no known relationships between the patients.

Are there recommended data splits (e.g., training, development/validation, testing)? If so, please provide a description of these splits, explaining the rationale behind them.

We provide training and validation splits, containing 531 (90%) and 59 (10%) of cases. These are derived through uniform sampling of the resulting dataset to ensure a balanced distribution of labels in both sets.

Are there any errors, sources of noise, or redundancies in the dataset? If so, please provide a description.

One possible error source for image annotation is uncertainty and inter-annotator variability. Prior literature suggests radiologist segmentation performance can vary 10-30% [14]. Another potential error source is image artifacts in CT that may result in suboptimal diagnostic quality.

Is the dataset self-contained, or does it link to or otherwise rely on external resources (e.g., websites, tweets, other datasets)? If it links to or relies on external resources, a) are there guarantees that they will exist, and remain constant, over time; b) are there official archival versions of the complete dataset (i.e., including the external resources as they existed at the time the dataset was created); c) are there any restrictions (e.g., licenses, fees) associated with any of the external resources that might apply to a dataset consumer? Please provide descriptions of all external resources and any restrictions associated with them, as well as links or other access points, as appropriate.

The dataset is self-contained.

Does the dataset contain data that might be considered confidential (e.g., data that is protected by legal privilege or by doctor-patient confidentiality, data that includes the content of individuals' non-public communications)? If so, please provide a description.

All data is anonymized and personally identifiable information has been removed. Please refer to each source dataset's publication for further details.

Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety? If so, please describe why.

No.

Does the dataset identify any subpopulations (e.g., by age, gender)? If so, please describe how these subpopulations are identified and provide a description of their respective distributions within the dataset.

No.

Is it possible to identify individuals (i.e., one or more natural persons), either directly or indirectly (i.e., in combination with other data) from the dataset? If so, please describe how.

No.

Does the dataset contain data that might be considered sensitive in any way (e.g., data that reveals race or ethnic origins, sexual orientations, religious beliefs, political opinions or union memberships, or locations; financial or health data; biometric or genetic data; forms of government identification, such as social security numbers; criminal history)? If so, please provide a description.

No.

Any other comments?

No.

2.3 Collection Process

How was the data associated with each instance acquired? Was the data directly observable (e.g., raw text, movie ratings), reported by subjects (e.g., survey responses), or indirectly inferred/derived from other data (e.g., part-of-speech tags, model-based guesses for age or language)? If the data was reported by subjects or indirectly inferred/derived from other data, was the data validated/verified? If so, please describe how.

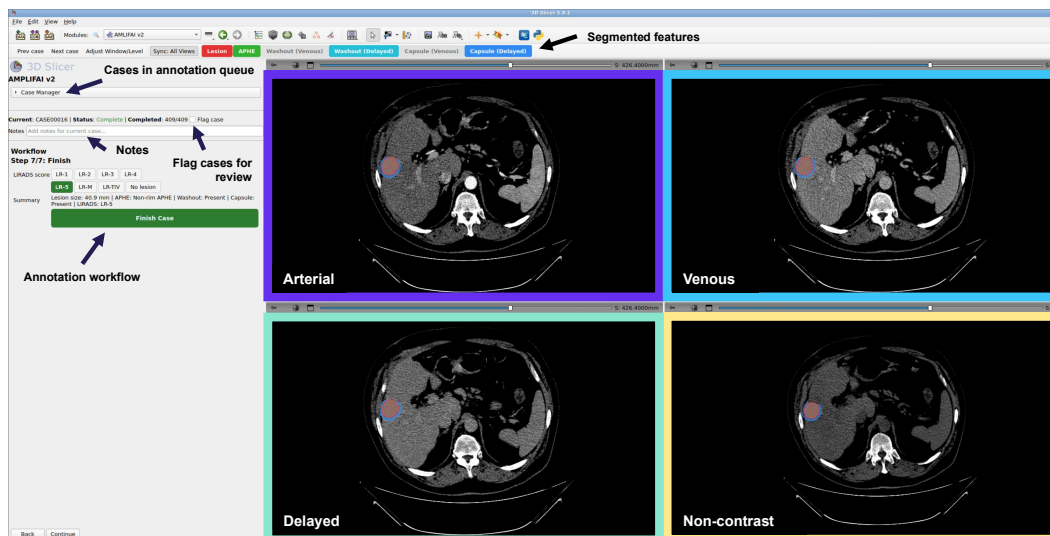


Figure 1: Overview of the radiologist annotation workflow. Annotators used a custom 3D Slicer interface to segment lesions and all three major LI-RADS on multiphase CT.

Annotation was performed by five board-certified radiologists and a radiology resident with substantial clinical experience in abdominal imaging. The radiologists were split into two groups: three attending annotators and two validators. The annotation team was onboarded to the annotation workflow, after which the same protocol was followed for all cases. Radiologists were blinded to the dataset splits, and no separate annotation procedures were used for different splits. Each case was independently annotated by the resident and one attending annotator. They were provided with annotations for liver lesions (if the dataset provided them) as well as AI-generated segmentation masks for each of the major LI-RADS features. To support an iterative annotation workflow, after manual segmentation of the first five cases was completed, a preliminary nnU-Net [15] was trained, and its predictions were manually refined if necessary. The nnU-Net was retrained after the review and refinement of 5, 20, 100, and 200 cases using an iterative learning approach, with a completely new model trained at each step to generate new predictions. The validators were not involved in the initial annotations, nor did they adjudicate or correct disagreements. Since all the imaging data is derived from four public datasets, please refer to each source dataset’s publication for further details.

What mechanisms or procedures were used to collect the data (e.g., hardware apparatuses or sensors, manual human curation, software programs, software APIs)? How were these mechanisms or procedures validated?

We developed a custom 3D slicer module to support annotation. It was hosted in a secure research environment at our hospital (Figure 1).

If the dataset is a sample from a larger set, what was the sampling strategy (e.g., deterministic, probabilistic with specific sampling probabilities)?

No.

Who was involved in the data collection process (e.g., students, crowdworkers, contractors) and how were they compensated (e.g., how much were crowdworkers paid)?

An interdisciplinary team from the Center for Applied AI at the University of Maryland Institute for Health Computing (UM-IHC) curated the dataset. The annotator team consisted of five board-certified radiologists and a radiology resident with substantial clinical experience in abdominal imaging: Jade Wong-You-Cheong, in practice since 1987 (39 years of experience); Florence X. Doo, in practice since 2018 (8 years of experience); Jeffrey D. Hirsch, in practice since 2001 (25 years of experience); Barton F. Lane, in practice since 2002 (24 years of experience); James Tonascia, in practice since 2020 (6 years of experience); and Joseph Chirico (4th year resident).

Over what timeframe was the data collected? Does this timeframe match the creation timeframe of the data associated with the instances (e.g., recent crawl of old news articles)? If not, please describe the timeframe in which the data associated with the instances was created.

The dataset consists of multiphase CT studies from four public datasets collected between November 2002 and December 2022. Please refer to each source dataset’s publication for further details. The annotations were performed between May 2026 and June 2026.

Were any ethical review processes conducted (e.g., by an institutional review board)? If so, please provide a description of these review processes, including the outcomes, as well as a link or other access point to any supporting documentation.

An IRB exemption was granted to conduct this retrospective study by the University of Maryland Institutional Review Board (HP-00117149).

Does the dataset relate to people? If not, you may skip the remaining questions in this section.

Yes, the dataset contains abdominal CT scans from human patients.

Did you collect the data from the individuals in question directly, or obtain it via third parties or other sources (e.g., websites)?

The dataset is derived from four public datasets: TCGA-LIHC [9], WAW-TACE [10], HCC-TACE-SEG [11], and PLC-CECT [12]. Please refer to each source dataset’s publication for further details.

Were the individuals in question notified about the data collection? If so, please describe (or show with screenshots or other information) how notice was provided, and provide a link or other access point to, or otherwise reproduce, the exact language of the notification itself.

Patients were not directly notified since the data is sourced from publicly available repositories. Please refer to each source dataset’s publication for further details.

Did the individuals in question consent to the collection and use of their data? If so, please describe (or show with screenshots or other information) how consent was requested and provided, and provide a link or other access point to, or otherwise reproduce, the exact language to which the individuals consented.

Please refer to each source dataset’s publication for further details.

If consent was obtained, were the consenting individuals provided with a mechanism to revoke their consent in the future or for certain uses? If so, please provide a description, as well as a link or other access point to the mechanism (if appropriate)

There is no mechanism to remove a patient from the dataset since the data is sourced from publicly available repositories. Please refer to each source dataset’s publication for further details.

Has an analysis of the potential impact of the dataset and its use on data subjects (e.g., a data protection impact analysis) been conducted? If so, please provide a description of this analysis, including the outcomes, as well as a link or other access point to any supporting documentation.

All data is anonymized and contains no personally identifiable information. Please refer to each source dataset’s publication for further details.

Any other comments?

No.

2.4 Preprocessing/Cleaning/Labeling

Was any preprocessing/cleaning/labeling of the data done (e.g., discretization or bucketing, tokenization, part-of-speech tagging, SIFT feature extraction, removal of instances, processing of missing values)? If so, please provide a description. If not, you may skip the remaining questions in this section.

The following preprocessing steps were applied to standardize the raw training data across all source datasets:

1. Removed irrelevant DICOM and DICOM-SEG series: TCGA-LIHC contained irrelevant series for our dataset (e.g., Scout, Dose Reports, etc.) which were removed. HCC-TACE-SEG contained combined lesion segmentations that were not compatible with our per-lesion annotation schema; these were removed.
2. Converted DICOM datasets to NIfTI volumes: TCGA-LIHC and HCC-TACE-SEG were provided in DICOM format and were converted to NIfTI using dcm2niix for standardization across the dataset.
3. Converted lesion segmentations from NRRD to NIfTI: WAW-TACE provided lesion segmentations in NRRD format; these were converted to NIfTI to maintain format consistency across all segmentation masks.
4. Phase Detection: We detected the contrast phase (non-contrast, arterial, portal venous, delayed) of each volume using imaging metadata, failing which, detection via temporal acquisition patterns using Comp2Comp [16, 17] was used. Phase labels were stored in the metadata of the dataset to enable phase-specific analysis and model development.
5. Image Registration: All multi-phase CT studies underwent image registration to align arterial, portal venous, and delayed phase acquisitions to a common spatial reference. The registration uses the venous phase, if available as the key image, if not then the delayed phase is utilized. Cases without a venous or delayed phase are invalid for our challenge. The preprocessing pipeline will be implemented centrally and applied uniformly to the training, validation, and test sets. The complete preprocessing code will be made publicly available to ensure transparency and reproducibility.

The preprocessing pipeline will be implemented centrally and applied uniformly to the training and validation sets. The complete preprocessing code will be made publicly available to ensure transparency and reproducibility.

Was the "raw" data saved in addition to the preprocessed/cleaned/labeled data (e.g., to support unanticipated future uses)? If so, please provide a link or other access point to the "raw" data.

The raw data is publicly accessible through each source dataset's repository. TCGA-LIHC and HCC-TAGE-SEG are both hosted on The Cancer Imaging Archive (TCIA), WAW-TACE is hosted on Zenodo, and PLC-CECT is hosted on ScienceDB.

Is the software that was used to preprocess/clean/label the data available? If so, please provide a link or other access point.

The complete preprocessing code will be made publicly available to ensure transparency and reproducibility.

Any other comments?

No.

2.5 Uses

Has the dataset been used for any tasks already? If so, please provide a description.

The dataset has been used to train several models for characterizing HCC lesions using the standardized LI-RADS criteria. These models take multiphase abdominal CT scans and a target lesion mask as input to predict the probability of its associated LI-RADS category (LR-1 through LR-5, LR-TIV, or LR-M). We explore two possible methods: First, an interpretable model may strictly follow the clinical criteria to predict all major LI-RADS features (tumor size, non-rim APHE, non-peripheral washout, and enhancing capsule) before deriving the final LI-RADS category. Second, an end-to-end "black-box" model may directly predict the LI-RADS category, using either all available contrast phases, or, in some cases, only a single phase.

Is there a repository that links to any or all papers or systems that use the dataset? If so, please provide a link or other access point.

We will include a partial list of papers that use the dataset on the [AMPLIFAI Challenge Website](#).

What (other) tasks could the dataset be used for?

Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses? For example, is there anything that a dataset consumer might need to know to avoid uses that could result in unfair treatment of individuals or groups (e.g., stereotyping, quality of service issues) or other risks or harms (e.g., legal risks, financial harms)? If so, please provide a description. Is there anything a dataset consumer could do to mitigate these risks or harms?

Are there tasks for which the dataset should not be used? If so, please provide a description.

The dataset should only be used for personal, non-commercial research. The dataset has neither been reviewed nor approved by the U.S. Food and Drug Administration (FDA) for clinical use. It should not be used for attempting to identify any individual nor diagnose patients with pathologies.

Any other comments?

No.

2.6 Distribution

Will the dataset be distributed to third parties outside of the entity (e.g., company, institution, organization) on behalf of which the dataset was created? If so, please provide a description.

The dataset is publicly available on the [AMPLIFAI Challenge Website](#).

How will the dataset will be distributed (e.g., tarball on website, API, GitHub)? Does the dataset have a digital object identifier (DOI)?

The dataset is distributed to challenge participants via [Hugging Face](#), containing all images, labels, and metadata for the training and validation splits. The total dataset size is about 146 GB.

When will the dataset be distributed?

The dataset is currently available.

Will the dataset be distributed under a copyright or other intellectual property (IP) license, and/or under applicable terms of use (ToU)? If so, please describe this license and/or ToU, and provide a link or other access point to, or otherwise reproduce, any relevant licensing terms or ToU, as well as any fees associated with these restrictions.

The dataset is released under [CC BY-NC-SA](#) license for personal, non-commercial research use.

Have any third parties imposed IP-based or other restrictions on the data associated with the instances? If so, please describe these restrictions, and provide a link or other access point to, or otherwise reproduce, any relevant licensing terms, as well as any fees associated with these restrictions.

No.

Do any export controls or other regulatory restrictions apply to the dataset or to individual instances? If so, please describe these restrictions, and provide a link or other access point to, or otherwise reproduce, any supporting documentation.

No. All data is sourced from publicly available repositories.

Any other comments?

No.

2.7 Maintenance

Who will be supporting/hosting/maintaining the dataset?

The dataset will be maintained by University of Maryland Institute for Health Computing.

How can the owner/curator/manager of the dataset be contacted (e.g., email address)?

Please contact Dr. Florence X. Doo via email: fdoo@som.umaryland.edu.

Is there an erratum? If so, please provide a link or other access point.

No.

Will the dataset be updated (e.g., to correct labeling errors, add new instances, delete instances)?

If so, please describe how often, by whom, and how updates will be communicated to dataset consumers (e.g., mailing list, GitHub)?

There are no plans to update the dataset at this time.

If the dataset relates to people, are there applicable limits on the retention of the data associated with the instances (e.g., were the individuals in question told that their data would be retained for a fixed period of time and then deleted)? If so, please describe these limits and explain how they will be enforced.

There are no limits on the retention of the information in the dataset.

Will older versions of the dataset continue to be supported/hosted/maintained? If so, please describe how. If not, please describe how its obsolescence will be communicated to dataset consumers.

No.

If others want to extend/augment/build on/contribute to the dataset, is there a mechanism for them to do so? If so, please provide a description. Will these contributions be validated/verified? If so, please describe how. If not, why not? Is there a process for communicating/distributing these contributions to dataset consumers? If so, please provide a description.

No. Since the data is released under **CC BY-NC-SA**, users are encouraged to extend, augment, or build on the dataset for non-commercial, research use with attribution.

Any other comments?

No.

References

- [1] Amit G Singal, Anjana Pillai, and Jasmin Tiro. Early detection, curative treatment, and survival rates for hepatocellular carcinoma surveillance in patients with cirrhosis: a meta-analysis. *PLoS medicine*, 11(4):e1001624, 2014.
- [2] Josep M Llovet, Robin Kate Kelley, Augusto Villanueva, Amit G Singal, Eli Pikarsky, Sasan Roayaie, Riccardo Lencioni, Kazuhiko Koike, Jessica Zucman-Rossi, and Richard S Finn. Hepatocellular carcinoma. *Nature Reviews. Disease Primers*, 10(1):10, 2024.
- [3] Victoria Chernyak, Kathryn J Fowler, Aya Kamaya, Ania Z Kielar, Khaled M Elsayes, Mustafa R Bashir, Yuko Kono, Richard K Do, Donald G Mitchell, Amit G Singal, et al. Liver imaging reporting and data system (li-rads) version 2018: imaging of hepatocellular carcinoma in at-risk patients. *Radiology*, 289(3):816–830, 2018.
- [4] Sunyoung Lee, Yeun-Yoon Kim, Jaeseung Shin, Won Jeong Son, Yun Ho Roh, Jin-Young Choi, Claude B Sirlin, and Victoria Chernyak. Percentages of hepatocellular carcinoma in li-rads categories with ct and mri: a systematic review and meta-analysis. *Radiology*, 307(1):e220646, 2023.
- [5] Stacy M Goins, Robert G Adamo, Eric Lam, Andreu F Costa, Christian B van der Pol, Jean-Paul Salameh, Haben Dawit, Matthew DF McInnes, and Mustafa R Bashir. Conversion strategy for li-rads category 5 observations across versions 2014, 2017, and 2018, 2023.
- [6] Stacy M Goins, Hanyu Jiang, Christian B Van der Pol, Jean-Paul Salameh, Eric Lam, Robert G Adamo, Matthew DF McInnes, Andreu F Costa, An Tang, Ayman S Alhasan, et al. Individual participant data meta-analysis of lr-5 in li-rads version 2018 versus revised li-rads for hepatocellular carcinoma diagnosis. *Radiology*, 309(3):e231656, 2023.
- [7] Robert G Adamo, Christian B van der Pol, Mostafa Alabousi, Eric Lam, Jean-Paul Salameh, Nicole Abedrabbo, Emily Lerner, Haresh Naringrekar, Mustafa R Bashir, Andreu F Costa, et al. Diagnostic performance of ct/mri li-rads version 2018 major feature combinations: individual participant data meta-analysis. *Radiology*, 315(3):e243450, 2025.

- [8] Hanning Ying, Xiaoqing Liu, Min Zhang, Yiyue Ren, Shihui Zhen, Xiaojie Wang, Bo Liu, Peng Hu, Lian Duan, Mingzhi Cai, et al. A multicenter clinical ai system study for detection and diagnosis of focal liver lesions. *Nature communications*, 15(1):1131, 2024.
- [9] Bradley Erickson, Shanah Kirk, Yueh Lee, Oliver Bathe, Melissa Kearns, Cindy Gerdes, Kimberly Rieger-Christ, and John Lemmerman. The cancer genome atlas liver hepatocellular carcinoma collection (tcga-lihc), 2016.
- [10] Krzysztof Bartnik, Tomasz Bartczak, Mateusz Krzyżiński, Krzysztof Korzeniowski, Krzysztof Lamparski, Piotr Węgrzyn, Eric Lam, Mateusz Bartkowiak, Tadeusz Wróblewski, Katarzyna Mech, et al. Waw-tace: A hepatocellular carcinoma multiphase ct dataset with segmentations, radiomics features, and clinical data. *Radiology: Artificial Intelligence*, 6(6):e240296, 2024.
- [11] Ahmed W Moawad, Ali Morshid, Ahmed M Khalaf, Mohab M Elmohr, John D Hazle, David Fuentes, Mohamed Badawy, Ahmed O Kaseb, Manal Hassan, Armeen Mahvash, et al. Multi-modality annotated hepatocellular carcinoma data set including pre-and post-tace with imaging segmentation. *Scientific data*, 10(1):33, 2023.
- [12] Jiawei Luo, Xiaoyu Wan, Jinchao Du, Li Liu, Ling Zhao, Xin Peng, Min Wu, Shixin Huang, and Xixi Nie. Comprehensive multi-phase 3d contrast-enhanced ct imaging for primary liver cancer. *Scientific Data*, 12(1):768, 2025.
- [13] Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé Iii, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021.
- [14] Iris Van Der Loo, Teresa M Tareco Bucho, James A Hanley, Regina GH Beets-Tan, Alex LT Imholz, and Stefano Trebeschi. Measurement variability of radiologists when measuring brain tumors. *European Journal of Radiology*, 183:111874, 2025.
- [15] Fabian Isensee, Paul F Jaeger, Simon AA Kohl, Jens Petersen, and Klaus H Maier-Hein. nnu-net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature methods*, 18(2):203–211, 2021.
- [16] Jakob Wasserthal, Hanns-Christian Breit, Manfred T Meyer, Maurice Pradella, Daniel Hinck, Alexander W Sauter, Tobias Heye, Daniel T Boll, Joshy Cyriac, Shan Yang, et al. Totalsegmentator: robust segmentation of 104 anatomic structures in ct images. *Radiology: Artificial Intelligence*, 5(5):e230024, 2023.
- [17] Louis Blankemeier, Arjun Desai, Juan Manuel Zambrano Chaves, Andrew Wentland, Sally Yao, Eduardo Reis, Malte Jensen, Bhanushree Bahl, Khushboo Arora, Bhavik N Patel, et al. Comp2comp: open-source body composition assessment on computed tomography. *arXiv preprint arXiv:2302.06568*, 2023.